5-3. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

(a) Do you think the engineer is correct? Use the P-value approach.

(b) If $\alpha = 0.05$, what is the power of the test in part (a) for a true difference in means of 0.04?

(c) Find a 95% CI on the difference in means. Provide a practical interpretation of this interval.

(d) Assuming equal sample sizes, what sample size should be used to ensure that $\beta = 0.01$ if the true difference in means is 0.04? Assume that $\alpha = 0.05$.

5-5. The burning rates of two different solid-fuel propellants used in aircrew escape systems are being studied. It is known that both propellants have approximately the same standard deviation of burning rate; that is, $\sigma_1 = \sigma_2 = 3$ cm/s. Two random samples of $n_1 = 20$ and $n_2 = 20$ specimens are tested; the sample mean burning rates are $\bar{x}_1 = 18.02$ cm/s and $\bar{x}_2 = 24.37$ cm/s.

(a) Test the hypothesis that both propellants have the same mean burning rate. Use a fixed-level test with $\alpha = 0.05$.

(b) What is the P-value of the test in part (a)?

(c) What is the β-error of the test in part (a) if the true difference in mean burning rate is 2.5 cm/s?

(d) Construct a 95% CI on the difference in means $\mu_1 - \mu_2$. What is the practical meaning of this interval?

5-9. Consider the situation described in Exercise 5-5. What ~~sample~~ size would be required in each population if we wanted the ~~error~~ in estimating the difference in mean burning rates to be less than 4 cm/s with ~~99%~~ confidence?

5-17. An article in *Electronic Components and Technology Conference* (Vol. 52, 2001, pp. 1167–1171) describes a study comparing single versus dual spindle saw processes for copper metallized wafers. A total of 15 devices of each type were measured for the width of the backside chipouts, $\bar{x}_{single} = 66.385$, $s_{single} = 7.895$ and $\bar{x}_{double} = 45.278$, $s_{double} = 8.612$.

(a) Do the sample data support the claim that both processes have the same chip outputs? Assume that both populations are normally distributed and have the same variance. Answer this question by finding and interpreting the P-value for the test.

(b) Construct a 95% two-sided confidence interval on the mean difference in spindle saw process. Compare this interval to the results in part (a).

(c) If the β-error of the test when the true difference in chip outputs is 15 should not exceed 0.1 when $\alpha = 0.05$, what ~~sample~~ sizes must be used?

5-24. Two suppliers manufacture a plastic gear used in a laser printer. The impact strength of these gears measured in foot-pounds is an important characteristic. A random sample of 10 gears from supplier 1 results in $\bar{x}_1 = 289.30$ and $s_1 = 22.5$, and another random sample of 16 gears from the second supplier results in $\bar{x}_2 = 321.50$ and $s_2 = 21$.

(a) Is there evidence to support the claim that supplier 2 provides gears with higher mean impact strength? Use ~~the~~ P-value approach, and assume that both populations are normally distributed but the variances are not equal.

(b) Do the data support the claim that the mean impact strength of gears from supplier 2 is at least 25 foot-pounds higher than that of supplier 1? Make the same assumptions as in part (a).

(c) Construct an appropriate ~~95%~~ CI on the difference in mean impact strength. Use this interval to answer the question posed in part (b).

5-41. A computer scientist is investigating the usefulness of two different design languages in improving programming tasks. Twelve expert programmers, familiar with both languages, are asked to code a standard function in both languages, and the time in minutes is recorded. The data are shown here:

	Time	
Programmer	Design Language 1	Design Language 2
1	17	18
2	16	14
3	21	19
4	14	11
5	18	23
6	24	21
7	16	10
8	14	13
9	21	19
10	23	24
11	13	15
12	18	20

(a) Find a 95% CI on the difference in mean coding times. Is there any indication that one design language is preferable?

(b) Is it reasonable to assume that the difference in coding time is normally distributed? Show evidence to support your answer.

5-59. Consider the gear impact strength data in Exercise 5-24. Is there sufficient evidence to conclude that the variance of impact strength is different for the two suppliers? Use $\alpha = 0.05$.

5-69. The rollover rate of sport utility vehicles is a transportation safety issue. Safety advocates claim that manufacturer A's vehicle has a higher rollover rate than that of manufacturer B. One hundred crashes for each of these vehicles were examined. The rollover ~~rates~~ were $\hat{p}_A = 0.35$ and $\hat{p}_B = 0.25$.

(a) Does manufacturer A's vehicle have a higher rollover rate than manufacturer B's? Use ~~the~~ P-value approach.

(b) What is ~~the~~ power of this test, assuming α ~~= 0.05~~?

(c) Assume that the rollover rate of manufacturer A's vehicle is 0.15 higher than B's. Is the sample size sufficient for detecting this difference with probability level at least 0.90, if α = ~~0.053~~?

5-71. Construct a ~~95%~~ lower bound on the difference in the two rollover rates for Exercise 5-69. Provide a practical interpretation of this interval.

5-73. Rework Exercise 5-71 using the new CI. Compare this interval to the traditional one.